

$$1. \quad S = (x, \theta, \dot{x}, \dot{\theta})$$

$$\dot{S} = f(S, u)$$

$$\bar{S} = (0, \pi, 0, 0)$$

$$S_{k+1} = S_k + \Delta t f(S_k, u_k)$$

$$\tilde{S}_k = S_k - \bar{S}$$

$$\tilde{S}_{k+1} \approx A \tilde{S}_k + B u_k$$

$$A = I + \Delta t \cdot \frac{\partial f}{\partial S}(\bar{S}, \bar{u})$$

$$= \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & \frac{m_p g}{m_c} \cdot \Delta t & 1 & 0 \\ 0 & \frac{(m_c + m_p)g}{m_c L} \cdot \Delta t & 0 & 1 \end{bmatrix}$$

$$B = \Delta t \cdot \frac{\partial f}{\partial u}(\bar{S}, \bar{u})$$

$$= \begin{bmatrix} 0 \\ 0 \\ \frac{\Delta t}{m_c} \\ \frac{\Delta t}{m_c L} \end{bmatrix}$$

4.

(a)

$$\therefore \bar{\theta} = \pi$$

$$\dot{\bar{\theta}} = 0$$

$$\bar{u} = 0$$

$\therefore$ dynamics doesn't depend on $\bar{x}(t)$
which means

$$A(t) = A$$

$$B(t) = B$$

do not need to change

(c)

The desired trajectory requires that
 $\dot{x}$ should decrease and increase
 θ should keep π
 $\dot{\theta}$ should always be 0

Q is the penalty weight of difference between
state and origin.

Increasing Q will decrease the error
of $x(t)$ and $\dot{x}(t)$ but doesn't
influence θ and $\dot{\theta}$